

❖ 综合诊疗 ❖ [Comprehensive Diagnosis and Treatment]

住院肿瘤患者潜在药物相互作用的发生率和严重程度及危险因素分析

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【摘要】 目的:调查分析住院肿瘤患者潜在药物相互作用(potential drug-drug interactions, pDDIs)的发生情况及其严重程度,并预测相关的危险因素。**方法:**抽取本院2022年03月至2022年07月住院肿瘤患者的医嘱,通过IBM Micromedex筛查潜在药物相互作用。**结果:**282例患者有137例发生pDDIs,发生率为48.58%。相互作用的类型主要为地塞米松磷酸钠和阿瑞匹坦、非甾体抗炎药及阿片类止痛药、卡培他滨和质子泵抑制剂。通过多因素Logistic回归分析发现,年龄(≥60岁)、手术史、体质指数(body mass index, BMI)、多药联用(≥5种)、住院天数(≥7天)等是发生pDDIs的危险因素。**结论:**肿瘤患者住院期间的pDDIs发生率高,临床医师及药师需要加强监测及干预相关危险因素,提高药物治疗的安全性和合理性。

【关键词】肿瘤;住院患者;潜在药物相互作用;危险因素

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Analysis of the incidence, severity and risk factors of potential drug interactions in hospitalized cancer patients

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【Abstract】 Objective: To investigate the prevalence and severity of potential drug-drug interactions (pDDIs) in hospitalized cancer patients, and predict the associated risk factors. **Methods:** Extract medical orders from hospitalized tumor patients in our hospital from March 2022 to July 2022, and screen for potential drug interactions through IBM Micromedex. **Results:** 137 of 282 patients had pDDIs, with an incidence of 48.58%. The main types of pDDIs were dexamethasone sodium phosphate and aprepitant, non-steroidal anti-inflammatory drugs and opioid analgesics, capecitabine and proton pump inhibitors. By multivariate Logistic regression analysis, age (≥60 years old), operation history, body mass index (BMI), polypharmacy (≥5 types), the length of stay in hospital (≥7 days) were the risk factors associated with pDDIs. **Conclusion:** The prevalence of pDDIs in hospitalized cancer patients is high, clinicians and pharmacists need to monitor and intervene the risk factors to improve the safety and rationality of drug treatment.

【Key words】 cancer, hospitalized patient, potential drug-drug interactions, risk factors

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在中国,癌症是一个重大的公共卫生问题。据中国国家癌症中心(China National Cancer Center, NCC)统计,2016年

中国新增癌症病例约4 064 000例,新增癌症死亡2 413 500例^[1]。由于肿瘤患者病情严重、治疗方案复杂,需要面对高风险的药物相互作用(drug-drug interactions, DDIs)^[2]。

DDIs是指一种药物在连续或同时给药时增强、减弱另一种药物的作用或效果的能力^[3]。药物相互作用是包括治疗失败在内的药物不良反应(adverse drug reaction, ADR)的重要原因^[4]。一项来自法国药物警戒数据库的分析显示,4 027份由药物相互作用引起的不良反应报告中,82.02%产生严重的ADR^[5]。然而此类不良反应本是可以预防和避免的^[6]。本研究通过调查肿瘤病人住院期间潜在药物相互作用(potential drug-drug interactions, pDDIs)的发生率、类型及明确相关危险因素,可以使临床医师、药师更好地识别、评估

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和预测潜在的药物相互作用,给个体化合理用药提供参考。

1 资料与方法

1.1 资料

本横断面研究纳入本院 2022 年 03 月至 2022 年 07 月住院肿瘤患者的医嘱。纳入患者标准:年龄 > 18 岁、住院时长 > 2 天、确诊为肿瘤患者、住院期间使用 2 种及其以上药物。药物医嘱、病人的基本特征等从医院 HIS 系统里获取。

1.2 方法

使用 IBM Micromedex 药物咨询数据库中的药物相互作用程序分析 pDDIs。药物相互作用严重性分为:禁忌 (contraindicated)、严重 (major)、中等 (moderate)、较弱 (minor),文献循证医学质量分为卓越 (excellent)、良好 (good)、一般 (fair)、未知 (unknown)。

1.3 统计分析

应用 IBM SPSS 25 软件对数据进行分析,计量资料采用均数 ± 标准差 ($\bar{x} \pm s$) 表示,简单关联性描述采用 χ^2 检验,危险因素采用 Logistic 回归分析, $P < 0.05$ 表示差异具有统计学意义。

2 结果

2.1 研究对象基本特征描述

共纳入 282 例患者,男性 155 例,占 55%,女性 127 例,占 45%,平均年龄 (60.55 ± 12.61) 岁,平均 KPS 评分 (82.11 ± 8.22) 分,平均服用药品 (9.52 ± 5.49) 种,平均住院天数为 (8.48 ± 7.26) 天。282 例患者中发生 pDDIs 有 137 例,占比 48.6%,具体见表 1。

表 1 患者的基本特征描述 $n(\%)$

Tab. 1 Basic characterization of patients $n(\%)$

Characteristics	Total population	95% CI
Age (years, $\bar{x} \pm s$)	60.55 ± 12.61	59.07 ~ 62.03
KPS rating ($\bar{x} \pm s$)	82.11 ± 8.22	81.15 ~ 83.07
Number of drug varieties ($\bar{x} \pm s$)	9.52 ± 5.49	8.88 ~ 10.16
Hospitalization days (d, $\bar{x} \pm s$)	8.48 ± 7.26	7.63 ~ 9.33
Gender		
Male	155 (55.0)	0.49 ~ 0.609
Female	127 (45.0)	
Marital status		
Unmarried	9 (3.2)	0.015 ~ 0.06
Married	273 (96.8)	
Surgical situation		
Unsurgical	80 (28.4)	0.232 ~ 0.34
Surgical	202 (71.6)	
Smoking or not		
Not smoking	185 (65.6)	0.597 ~ 0.711
Smoking	97 (34.4)	
Whether to drink alcohol or not		
Not drinking alcohol	213 (75.5)	0.701 ~ 0.804
Drinking	69 (24.5)	
Allergic situation		
Not allergic	264 (93.6)	0.901 ~ 0.962
Allergy	18 (6.4)	
Whether there is a family history		
No family history	255 (90.4)	0.864 ~ 0.936
Family history	27 (9.6)	
Has pDDIs occurred		
Nothing	145 (51.4)	0.454 ~ 0.574
Yes	137 (48.6)	

2.2 发生 pDDIs 情况

经过审核所有患者用药医嘱后,发生 pDDIs 有 137 例患者,共 290 条药品潜在相互作用,严重程度主要为 Major 和 Moderate,分别占 60.00%、36.20%,见表 2。

表 2 pDDIs 中严重程度数量及比例

Tab. 2 Number and proportion of severity in pDDIs

Severity	Quantity	Proportion (%)
Contraindicated	4	1.38
Major	174	60.00
Moderate	105	36.20
Minor	7	2.42

经过系统梳理,290 条 pDDIs 中,占前 8 位的两药相互作用类型如下表 3,排名前 4 位分别为地塞米松磷酸钠和阿瑞匹坦、地塞米松磷酸钠和非甾体抗炎药、地塞米松磷酸钠和阿片类止痛药、卡培他滨和质子泵抑制剂。

2.3 简单关联性描述

本研究共纳入研究对象 282 例。结局按照有或无 pDDIs 发生分为两组。两组总体多药联用差异有统计学意义 ($\chi^2 = 49.711, P = 0.000$),两组总体住院天数差异有统计学意义 ($\chi^2 = 19.208, P = 0.000$),两组总体 BMI 差异有统计学意义 ($\chi^2 = 6.632, P = 0.035$)。住院多药联用 (≥ 5 种)、住院天数 (≥ 7 天)、BMI 与是否发生 pDDIs 呈显著性相关。其他性别、年龄、婚姻、手术、吸烟、饮酒、过敏史、家族史癌症类型无统计学差异,详见表 4。

2.4 多因素 Logistic 回归分析

纳入年龄组、婚姻状况和是否手术、吸烟等构建 Logistic 回归方程,见表 5。Omnibus 检验、拟合优度检验提示模型总体有意义、拟合效果较好。结果发现:年龄 (≥ 60 岁)、手术史、BMI ($> 18.5 \text{ kg/m}^2$)、药品品种 (≥ 5 种)、住院天数 (≥ 7 天)是发生 pDDIs 的危险因素。

3 讨论

本研究住院肿瘤患者 pDDIs 发生率 48.6%,介于口服抗癌药物治疗发生率 16%^[7]和静脉注射治疗的非卧床癌症患者发生率 58%^[8]之间,另一项加拿大研究报告门诊肿瘤患者 pDDIs 发生率 27%^[9],这些差异性可能与纳入和排除标准、研究人群及其特征、就诊模式、药物治疗途径、临床药学服务的存在与否等有关。

本研究中发生 pDDIs 频次最高的药物是地塞米松磷酸钠。在肿瘤科,地塞米松磷酸钠应用广泛,是一种长效而廉价的糖皮质激素,提高肿瘤治疗效果,减少化疗引起的毒副反应^[10]。排名第一的 pDDIs 为阿瑞匹坦与地塞米松磷酸钠,同时使用可能导致地塞米松暴露增加。地塞米松代谢主要依赖 CYP3A 酶家族^[11-12],有研究发现,阿瑞匹坦是 CYP3A4 弱抑制药^[13],抑制地塞米松肝代谢,增加激素作用。当与口服阿瑞匹坦方案同时给予成人时,地塞米松剂量应在第 1 日减少至 12 mg,对于 12 岁或以上的儿科患者,建议地塞米松剂量应减少 50%^[14]。

排名第二的 pDDIs 为地塞米松磷酸钠与非甾体抗炎药。同时使用可能会增加胃肠道溃疡或出血的风险。根据危险程度,往往需要增加质子泵抑制剂来进行干预^[15]。一些肿瘤患者会使用氟喹诺酮类药物,可能会导致肌腱炎和肌腱断

裂的风险,如果同时服用地塞米松磷酸钠,且年龄在 60 岁以上,肌腱断裂的风险尤其增加^[16-18],如果出现肌腱肿痛、断裂,应立即停止服用氟喹诺酮类。

表 3 前 8 位 pDDIs 的类型

Tab. 3 Top 8 types in pDDIs

Drug A	Drug B	Frequency	Level	Potential impact
Dexamethasone sodium phosphate	Aprepitant	74	Moderate	Increased exposure to dexamethasone sodium phosphate
	Celecoxib			
Dexamethasone sodium phosphate	Ibuprofen	47	Major	Increase the risk of gastrointestinal ulcers or bleeding
	Ketorolac butanediol			
	Diclofenac sodium			
Dexamethasone sodium phosphate	Oxycodone	19	Major	Reduce opioid efficacy or cause withdrawal syndrome
	Tramadol			
Capecitabine	Rabeprazole	17	Major	Reduced efficacy of capecitabine
	Omeprazole			
Dexamethasone sodium phosphate	Moxifloxacin	5	Major	Increasing the risk of tendon rupture
	Levofloxacin			
Potassium chloride	Spironolactone	4	Major	Increasing the risk of hyperkalemia
Dexamethasone sodium phosphate	Rivaroxaban	3	Moderate	Increased risk of gastrointestinal bleeding
Methoxyclopramide	Oxycodone	3	Major	Increasing the risk of central nervous system depression

表 4 pDDIs 影响因素差异性分析 $n(\%)$ Tab. 4 Difference analysis of influencing factors in pDDIs $n(\%)$

Variable grouping	Non occurring group	Occurring group	Chi-square test	
			χ^2	P
Gender			0.785	0.403
Female	69(54.3)	58(45.7)		
Male	76(49.0)	79(51.0)		
Age group(years)			0.928	0.343
<60	70(48.6)	74(51.4)		
≥60	75(54.3)	63(45.7)		
Marital status			0.181	0.744
Unmarried	4(44.4)	5(55.6)		
Married	141(51.6)	132(48.4)		
With or without surgery			0.574	0.509
Unsururgical	44(55.0)	36(45.0)		
With surgery	101(50.0)	101(50.0)		
Smoking or not			0.520	0.531
Not smoking	98(53.0)	87(47.0)		
Smoking	47(48.5)	50(51.5)		
Whether to drink alcohol or not			0.930	0.406
Not drinking alcohol	113(53.1)	100(46.9)		
Drinking alcohol	32(46.4)	37(53.6)		
Have any allergies			0.374	0.629
Not allergic	137(51.9)	127(48.1)		
Allergy	8(44.4)	10(55.6)		
Whether there is a family history			0.735	0.424
No family history	129(50.6)	126(49.4)		
Family history	16(59.3)	11(40.7)		
Number of combined drug varieties			49.711	0.000
<5	49(96.1)	2(3.9)		
≥5	96(41.6)	135(58.4)		
Hospitalization days(d)			19.208	0.000
<7	99(63.1)	58(36.9)		
≥7	46(36.8)	79(63.2)		
Cancer type			1.641	0.245
Solid tumor	141(52.2)	129(47.8)		
Hematological tumors	4(33.3)	8(66.7)		
BMI(kg/m ²)			6.632	0.035
<18.5	26(70.30)	11(29.7)		
18.5 ~ 23.9	69(50.7)	67(49.3)		
≥24	50(45.9)	59(54.1)		

表 5 pDDIs 的多因素 Logistic 回归分析
Tab. 5 Multivariate Logistic regression analysis of pDDIs

Variable grouping	B - value	B - value standard error	Wald chi square value	P	OR	OR 95.0% CI	
						Lower limit	Upper limit
Gender group							
Female *							
Male	-0.044	0.368	0.014	0.905	0.957	0.45	1.971
Age group(years)							
<60 *							
≥60	-0.652	0.296	4.855	0.028	0.521	0.292	0.930
Marital status							
Unmarried *							
Married	-0.754	0.878	0.738	0.390	0.471	0.084	2.628
Surgery or not							
Unsurgical *							
Operation	0.793	0.338	5.503	0.019	2.210	1.139	4.286
Smoking or not							
Not smoking *							
Smoke	0.367	0.419	0.769	0.380	1.444	0.635	3.281
Whether to drink alcohol or not							
Not drinking alcohol *							
Drink wine	0.213	0.402	0.281	0.596	1.237	0.563	2.719
History of drug allergies							
No allergies *							
Having allergies	0.445	0.583	0.583	0.445	1.561	0.498	4.895
Family history							
None *							
Yes	-0.784	0.473	2.747	0.097	0.456	0.1820	1.154
BMI(kg/m ²)							
<18.5 *							
18.5 ~ 23.9							
≥24	0.543	0.222	5.999	0.014	1.722	1.115	2.660
Number of combined drug varieties							
<5 *							
≥5	3.457	0.757	20.860	0.000	31.729	7.197	139.886
Hospitalization days(d)							
<7 *							
≥7	0.942	0.310	9.226	0.002	2.565	1.397	4.711
Tumor type							
Solid tumor *							
Hematological tumors	0.711	0.781	0.829	0.362	2.036	0.441	9.411

注: * 对照组。

Note: * Control group.

抗肿瘤药卡培他滨与质子泵抑制剂同时使用可能导致卡培他滨疗效降低。在 TRIO - 013/LOGIC 试验的特别分析中,在接受卡培他滨/奥沙利铂治疗的患者($n=117$)中,未同时使用 PPI 的患者与同时使用 PPI 相比,中位无进展生存期(PFS, 5.7 vs 4.2 个月)和中位总生存期(OS, 11.3 vs 9.2 个月)显著延长^[19],在一项回顾研究早期结直肠癌患者的实验中($n=298$),在任何时候,卡培他滨单一疗法与 PPI 联合使用与复发风险增加相关^[20]。

本研究中,年龄(≥ 60 岁)是肿瘤患者发生 pDDIs 的危险因素,老年肿瘤患者通常有不同的基础疾病,导致相关的生理状态发生改变,引起 pDDIs 的风险进一步增加^[21-22]。同时多药联用(≥ 5 种)和长时间住院(≥ 7 天)的肿瘤患者发生 pDDIs 的风险也会更高,这与郑灵招等^[23]、朱福来^[24]、

ALLAN RAMOS - ESQUIVEL 等^[2]的结果相一致,有研究表明,服用五种药物的患者暴露于 pDDI 的风险约为 50%,服用七种药物的患者接近 100%^[25-26]。癌症患者通常需要接受大量药物治疗,除了抗癌药物还包括支持性护理药物以及合并症药物等,通过本研究有助于临床医师了解肿瘤患者 pDDI 的类型及严重程度,提高对 pDDI 风险的认识,在抗癌药物治疗期间采取有效干预措施来预防 pDDI 的发生,有助于促进合理用药,从而改善患者的治疗结果和生活质量。

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❖ 肿瘤影像及介入 ❖ [Tumor Imaging and Image - directed Interventions]

鼻咽癌放疗后认知功能损害的静息态功能磁共振成像研究

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【摘要】 目的: 采用静息态脑功能成像技术检查鼻咽癌患者放射治疗后早期脑部功能损伤。方法: 对 9 例鼻咽癌患者放射治疗后及 9 例鼻咽癌患者放射治疗前进行脑部静息态功能成像, 放射治疗前与放射治疗后的患者为非同源对照, 采用成组 *t* 检验方法统计 2 组患者间全脑局部一致性 (regional homogeneity, ReHo)、低频振幅 (amplitude of low frequency fluctuation, ALFF) 的差异。结果: 鼻咽癌放射治疗后患者出现左侧小脑半球及双侧小脑脚 ReHo 减低; 并出现右侧额中回、双侧额下回 ALFF 值降低。结论: 鼻咽癌放射治疗后患者在静息状态下存在异常的脑部神经活动, 这些异常的脑部神经活动可能和放射治疗后认知功能障碍相关。

【关键词】 鼻咽癌; 放射治疗; 功能磁共振成像; 静息态

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Detection of cognitive impairments after radiotherapy for patients with nasopharyngeal carcinoma using resting - state brain functional magnetic resonance imaging

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【Abstract】 Objective: To detection of cognitive impairments after radiotherapy for patients with nasopharyngeal carcinoma using resting - state brain functional magnetic resonance imaging. **Methods:** 9 patients with nasopharyngeal carcinoma after radiotherapy and 9 patients with nasopharyngeal carcinoma before radiotherapy were scanned using resting - state brain functional magnetic resonance imaging, the differences of regional homogeneity (ReHo) and amplitude of low frequency fluctuation (ALFF) between the two groups were examined. Voxel based analysis was performed using two - sample *t* - test. **Results:** In patients with nasopharyngeal carcinoma after radiotherapy, decreased ReHo was detected in left cerebellum and bilateral cerebellar peduncles. Also, decreased ALFF was found in the right middle frontal gyrus, bilateral inferior frontal gyrus. No increase in ReHo or ALFF was found in the brain area of patients with nasopharyngeal carcinoma after radiotherapy. **Conclusion:** Patients with nasopharyngeal carcinoma after radiotherapy show abnormal brain nerve activity at resting state, which may be related to cognitive dysfunction after radiotherapy.

【Key words】 nasopharyngeal carcinoma, radiotherapy, functional magnetic resonance imaging, resting state

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